

**Knowing Eye: A Web-Based Examination Platform with Behavior Monitoring
Using Facial and Postural Analysis**



LEGACY COLLEGE OF COMPOSTELA

Dagohoy Street, Poblacion, Compostela, Davao de Oro

Institute of Information Technology

A Proposed Project Title Submitted by:

Saturnino C. Ancog III

Khrisha Marie O. Cavan

Kervy N. Cadiente

Twixt Jasley J. Tamera

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INTRODUCTION

Background of the Study

The use of digital technologies has changed how institutions conduct examinations, certifications, and competency assessments. Online and computer-based examination systems are now widely used by educational institutions, government agencies, and training organizations because they offer efficiency, scalability, and accessibility. However, as examinations shift to digital platforms, maintaining integrity, fairness, and reliability has become a major concern. Unlike traditional in-person exams, online assessments make it more difficult to monitor examinee behavior and prevent dishonest practices.

Globally, web-based examination systems support automated test administration, centralized data management, and remote access. However, existing studies identify one of the main limitations of online assessments as the difficulty of ensuring academic integrity when supervision is conducted remotely. Human proctors cannot effectively monitor a large number of examinees at the same time, especially in virtual environments. Because of this, researchers have explored the use of artificial intelligence and computer vision to improve monitoring through automated detection of behavior and identity verification.

Recent studies show the growing use of AI-based proctoring systems in online examinations. The iExam system [1] utilizes face detection and recognition to monitor students, achieving 90.4% face detection accuracy and 98.4% face recognition accuracy while detecting abnormal behaviors such as face disappearance and identity

changes. Similarly, the AutoOEP framework [2] combines multiple techniques such as gaze tracking, head pose estimation, and object detection to compute cheating probability in real time. These studies demonstrate a shift from simple identity verification to continuous behavioral monitoring using multiple visual indicators.

Other studies further support the use of automated proctoring. The Talent-Interview system [3] performs cheating detection directly on the client side using face recognition, object detection, and voice monitoring. In addition, AI-based proctoring research [4] shows that systems can automatically flag suspicious behavior and record evidence for review by human proctors. Another study reports 94.2% facial verification accuracy and 89.7% head movement detection accuracy using computer vision techniques [5]. These findings suggest that AI can improve monitoring efficiency while assisting human supervision rather than replacing it.

Advancements in computer vision models have also improved real-time monitoring capabilities. YOLO-based systems [6] enable object detection models to identify facial orientation, mouth movement, and prohibited objects such as mobile phones during exams. Furthermore, multi-perspective monitoring approaches [7] show that using multiple cameras and gaze recognition improves detection accuracy and provides better analysis of examinee behavior. These developments indicate that modern systems focus not only on presence detection but also on understanding behavioral patterns

Despite these advancements, many institutions still use partially integrated systems. Based on environmental scanning, Legacy College of Compostela (LCC) currently does not utilize a computer-based examination system with integrated face

tracking and identity verification. However, detection of suspicious behavior still depends on CCTV monitoring or assigned proctors. During large-scale examinations, this approach makes it difficult to monitor all examinees consistently, which may lead to missed suspicious behaviors and increased workload for proctors.

In addition, many existing systems focus mainly on exam administration, such as question management and automated scoring, while providing only passive monitoring features like webcam recording or screen locking. These systems do not analyze behavior in real time, limiting their ability to detect actions related to gaze direction, posture, or facial movement. As a result, institutions still rely heavily on manual review processes.

Given these limitations, there is a need for an integrated system that combines examination management and automated behavioral monitoring. Existing solutions are often fragmented, separating exam delivery from monitoring tools, which reduces efficiency and limits analytical support.

To address this gap, this study proposes Knowing Eye: A Web-Based Examination Platform with Behavior Monitoring Using Facial and Postural Analysis. The system integrates exam creation, administration, and monitoring into a single platform. It uses computer vision and deep learning techniques such as YOLO for detection, convolutional neural networks (CNN) for feature extraction, and FaceNet or ArcFace for facial verification. The system analyzes head movement, eye gaze, and posture in real time and generates behavioral reports to assist administrators in evaluating exam integrity.

This study supports the Commission on Higher Education (CHED) ACHIEVE Framework by promoting digital innovation and the use of emerging technologies to address real-world challenges. By integrating artificial intelligence and web-based systems, the study aims to improve efficiency, transparency, and reliability in modern examination environments.

For clarity, key terms are defined in this study. Online examination platforms refer to systems used to create and manage exams digitally. Computer vision refers to techniques that allow computers to interpret visual data from images or video. Behavioral monitoring refers to the analysis of examinee actions such as gaze, posture, and facial movement. Deep learning refers to neural network-based methods used for pattern recognition and classification.

Overall, this study responds to the need for more reliable and efficient online examination systems. By combining automated behavioral monitoring with exam management, the proposed system aims to reduce reliance on manual supervision and support fair and consistent evaluation processes.

Objectives of the Study

General Objectives

To develop a Knowing Eye: A Web-Based Examination Platform with Behavior Monitoring Using Facial and Postural Analysis to enhance examination integrity, monitoring efficiency, and assessment reliability.

Specific Objectives

1. Develop a centralized web-based examination platform capable of managing exam creation, administration, and monitoring within a unified system.
2. Develop a monitoring and data management system that captures and stores examination-related data.
3. Apply computer vision techniques to analyze examinee behavior in real time in order to:

3.1 Detect facial presence and identity consistency

3.2 Track head movement, eye gaze, and posture

3.3 Identify abnormal or suspicious behavioral patterns

4. Integrate deep learning algorithms for behavioral analysis by:

4.1 Utilizing YOLO for face and posture detection

4.2 Applying convolutional neural networks (CNN) for feature extraction

4.3 Implementing FaceNet or ArcFace for facial feature consistency verification

5. Develop a behavior monitoring and scoring module that:

5.1 Flags potential anomalies during examination sessions

5.2 Generates behavioral indicators based on detected patterns

5.3 Supports review of flagged events by administrators

6. Generate an administrative dashboard that visualizes examination data and monitoring results to assist administrators in:

6.1 Reviewing examination sessions

6.2 Analyzing behavioral reports

6.3 Supporting decision-making regarding examination integrity

7. Evaluate the functionality, usability, and effectiveness of the developed platform in improving examination monitoring efficiency and supporting fair and reliable assessment processes.

Significance of the Study

This section presents the significance of the study by identifying its beneficiaries and explaining how the proposed system may contribute to improving examination:

Educational Institution. The proposed system helps improve examination integrity, monitoring efficiency, and assessment reliability through automated behavior monitoring and centralized examination management.

Examination Proctors. The system assists in monitoring examinees by detecting and recording suspicious behaviors, reducing the workload associated with manual supervision.

Instructors. The system provides additional behavioral reports that support fair and transparent evaluation of examination sessions.

Students. The system promotes a secure and fair examination environment by ensuring consistent monitoring during assessments.

Future Researchers. This study may serve as a reference for future researchers in creating web-based examination systems that utilize computer vision, artificial intelligence, and behavior monitoring technologies.

Scope and Limitation

Scope

The development of Knowing Eye: A Web-Based Examination Platform with Behavior Monitoring Using Facial and Postural Analysis. It aims to provide a centralized platform that enables examination creation, administration, and automated behavioral monitoring.

This study is conducted at Legacy College of Compostela (LCC) located on Dagohoy St. Poblacion, Compostela, Davao de Oro. Moreover, data utilized in this study consist of video streams captured through examinees' web cameras during examination sessions, system-generated behavioral logs, and examination interaction records. Publicly available datasets are used to support facial and posture detection

model training, while additional controlled datasets are collected through recorded mock examination sessions with participant consent. These datasets are used solely for model training, testing, and performance evaluation purposes.

Limitation

The study is limited for institutional examination environments where online or web-based implementation using desktop or laptop equipped with webcams. Mobile application deployment, integration with external institutional databases, biometric hardware devices, and nationwide operational deployment are beyond the scope of this research. Additionally, the system has no offline capabilities and It provides behavioral indicators rather than definitive accusations of cheating, with final judgment remaining under administrator discretion.

Review of Related Literature and Works

Synthesis

Definition of Terms

CHAPTER 2

METHODOLOGY

This chapter presents the methodology utilized in the development of the proposed system entitled “Knowing Eye: A Web-Based Examination Platform with Behavior Monitoring Using Facial and Postural Analysis.” The study adopted the Agile Software Development Methodology as the framework for system development. Agile is a lightweight and adaptive software development approach that promotes continuous improvement, stakeholder collaboration, iterative development, and rapid response to changing requirements. Unlike traditional development methods, Agile focuses on delivering functional software in incremental stages while allowing modifications throughout the development process[#].

Figure # illustrates the Agile Software Development Methodology adopted in this study. The methodology follows the major phases of the System Development Life Cycle (SDLC), namely Planning, Design, Development, Testing, Deployment, and Maintenance. Each phase contributes to the successful development and continuous improvement of the proposed examination platform.

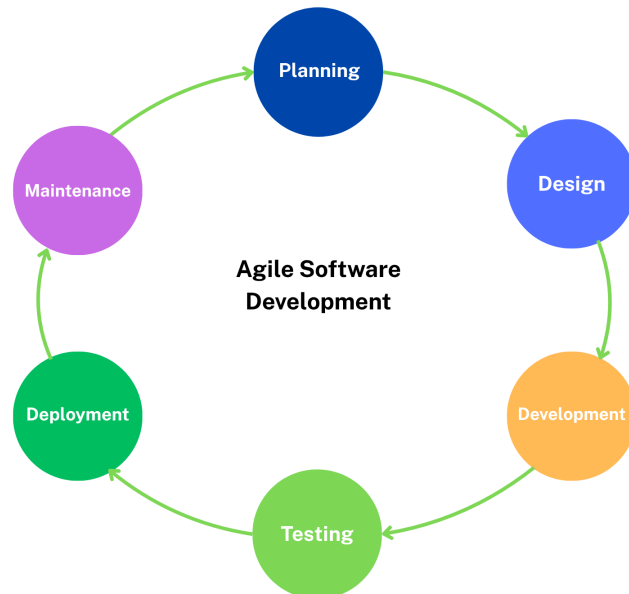


Figure #. Agile Software Development Methodology

System Planning

Bla bla

Project Team Organization

Each member of the project development team is assigned specific roles and responsibilities that contribute to the successful completion of the project.

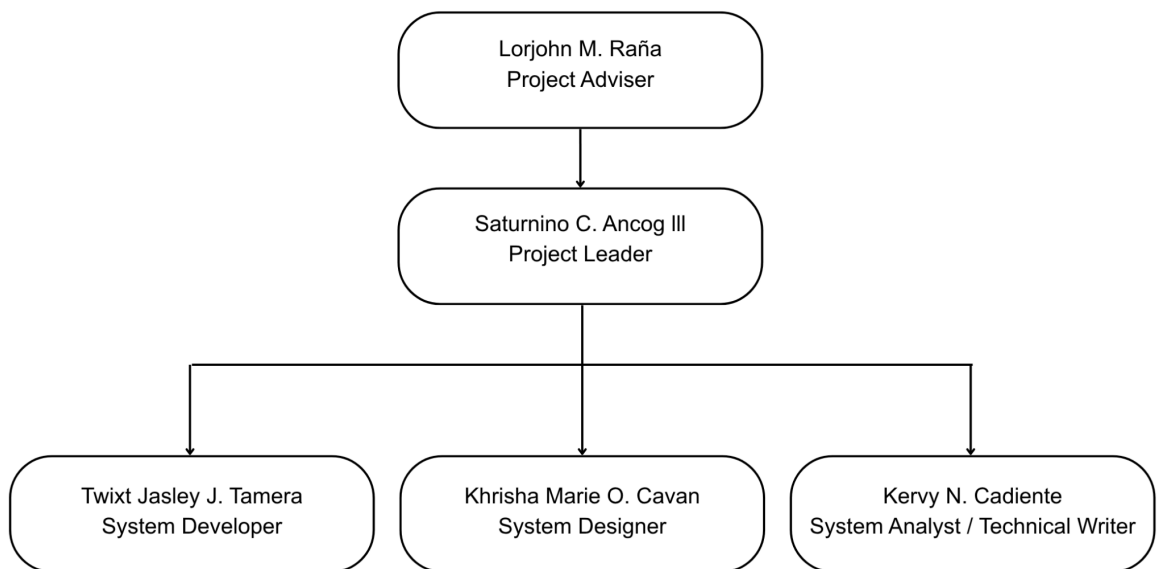
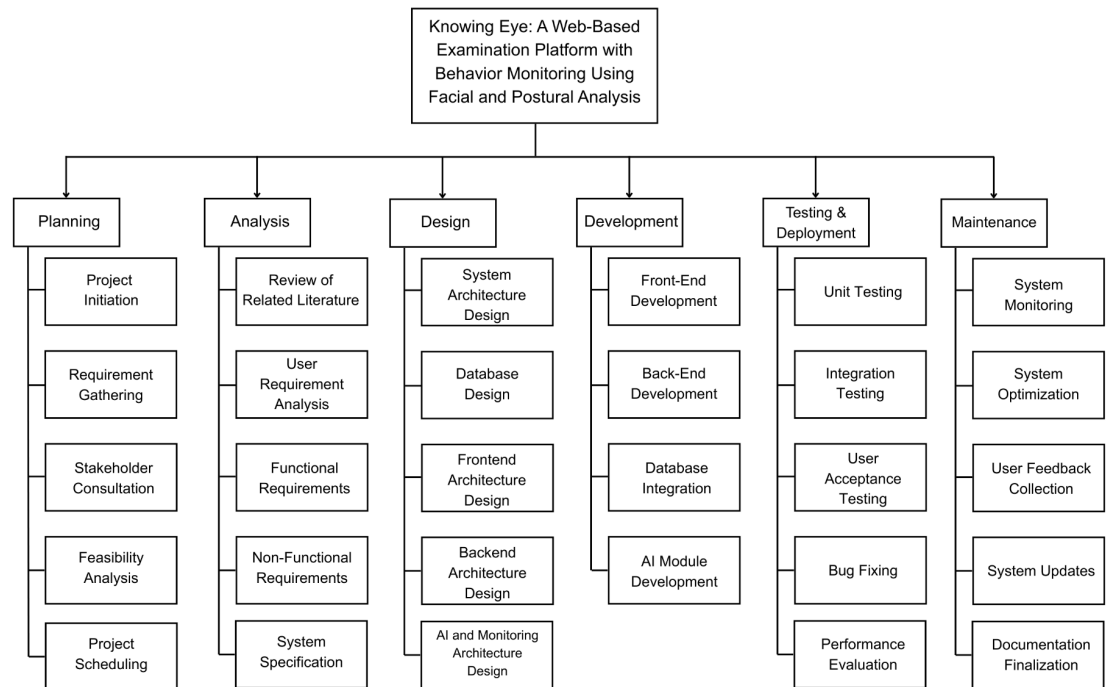


Figure #. Chart of the Project Development Team

Work Breakdown Structure



The project follows the PADIM-based development framework consisting of Planning, Analysis, Design, Implementation, and Maintenance phases. Each phase contains specific activities necessary for the completion of the proposed system.

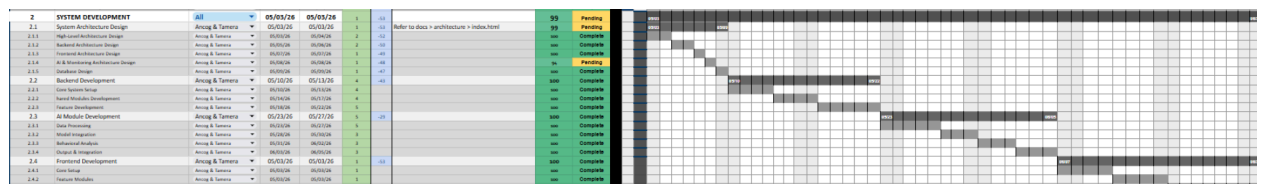
Figure #. Work Breakdown Structure

Gantt Chart

This section presents the project schedule for the development of Knowing Eye: A Web-Based Examination Platform with Behavior Monitoring Using Facial and Postural Analysis, as listed in the Work Breakdown Structure. A Gantt Chart is a graphical representation of the project timeline that illustrates the duration and sequence of activities throughout the development process. Figure # shows the schedule of major project activities from planning to maintenance.



[label figure as project planning]



[label figure as system development]

Figure #. Gantt Chart of the Project

System Analysis

After conducting environmental scanning, the researchers identified the requirements needed for the development of the proposed system. The system is designed to provide an online examination platform with real-time behavior monitoring through facial and postural analysis. It enables examinees to take examinations online while allowing administrators to manage examinations and monitor user activities.

System Architecture

Figure # presents the system architecture of the proposed web-based examination platform. The system consists of the Client Tier, Server Tier, and Data Tier.

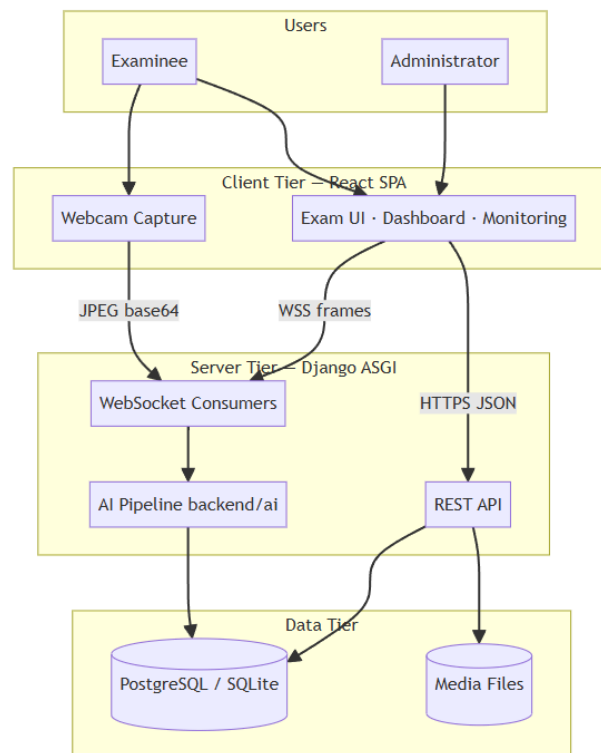


Figure # : Conceptual System Architecture

Functional and Non-Functional Requirements

Requirements were derived from capstone objectives, stakeholder interviews with IT faculty, and iterative prototype validation. Each item is numbered for traceability to test cases (prefixes FR-, NFR-).

Function Requirements

1. The system shall allow users to register and authenticate using username/password with JWT access and refresh tokens.
2. The system shall enforce role-based access control with two roles: Administrator and Examinee.
3. Administrators shall create, update, publish, archive, and delete exams and associated questions.
4. Administrators shall manage user accounts (activate, deactivate, assign roles).
5. Examinees shall start an exam session for a published exam within its availability window and attempt limit.
6. The system shall enforce one active session per examinee per exam at a time.
7. Examinees shall answer questions within a timed session and submit responses for automatic scoring.
8. The system shall calculate percentage score and pass/fail based on exam passing threshold.
9. Examinees shall enroll a reference face image before or during monitoring for identity verification.

10. The system shall capture webcam frames and analyze behavior in near real time via WebSocket (REST fallback).
11. The AI pipeline shall detect face absence, multiple faces, gaze deviation, poor posture, prohibited objects, and identity mismatch.
12. The system shall persist behavior events and generate severity-classified alerts linked to the session.
13. Administrators shall view live alerts and resolve them through the behavior API.
14. Administrators shall access dashboard KPIs, session reports, timeseries analytics, and CSV export.
15. Examinees shall view their own session history and results where authorized.

Non-Function Requirements

1. Performance: Frame analysis target ≤ 300 ms per frame at $\sim 1\text{--}5$ FPS; REST API responses ≤ 2 s under normal load.
2. Scalability: Support multiple concurrent examinees on a single server; production deployment uses Redis channel layer for multi-worker WebSocket broadcast.
3. Security: Passwords hashed (Django PBKDF2); JWT for API/WS auth; HTTPS/WSS in production; RBAC on all endpoints.
4. Reliability: AI adapter degrades to deterministic stub when ML dependencies unavailable; monitoring API contract always honored.
5. Usability: Responsive web UI; role-specific navigation; clear exam timer and compliance feedback during monitoring.

6. Maintainability: Feature-based modular monolith; documented REST catalog; 37 automated backend tests.
7. Portability: Runs on Windows/Linux; SQLite for development; PostgreSQL for production without code changes.
8. Privacy: Raw video frames are not stored in the database by default (KE_STORE_FRAMES=False); only derived metrics and events persist.

Use Case Diagram

A Use Case Diagram presents the interactions between the users and the proposed Knowing Eye System. As shown in Figure #, the system has two main actors: the Administrator and the Examinee.

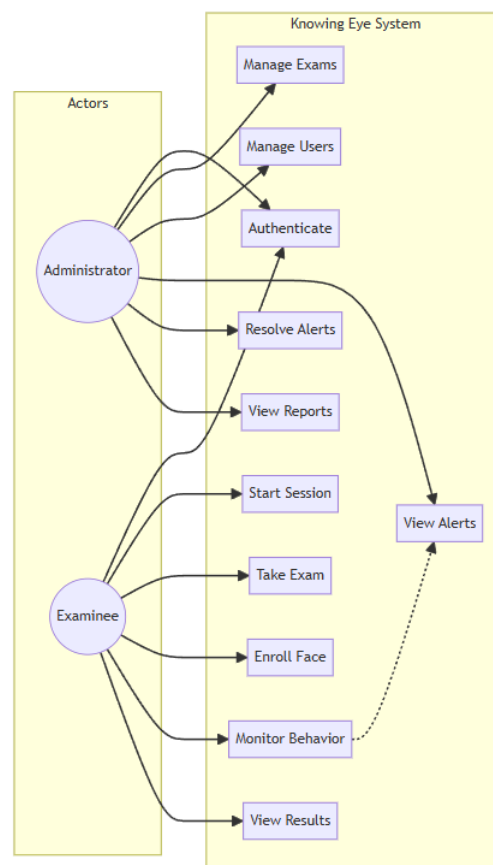


Figure #: Use Case Diagram

Context Flow Diagram (Level 0 DFD)

The researchers have used the Context Flow Diagram (CFD) to show the data exchange between the Administrator, Examinee, Webcam Device, and the Knowing Eye Examination Platform. Figure 10 presents the inputs and outputs of the system, including exam data, credentials, answers, video frames, alerts, reports, scores, and feedback.

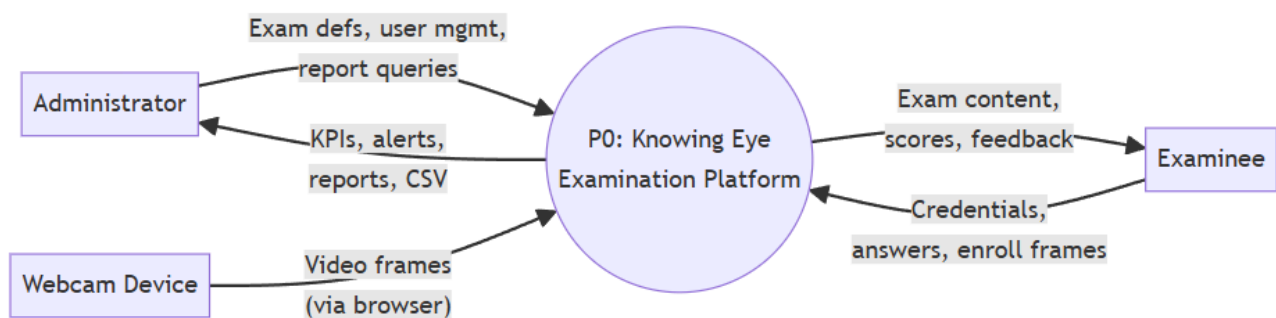


Figure #: ERD Level 0

Level 1 DFD

The researchers have used the Level 1 Data Flow Diagram (DFD) to show the detailed processes of the system. Figure 11 presents authentication, exam management, exam sessions, behavior analysis, alert recording, and report generation, including the related data stores.

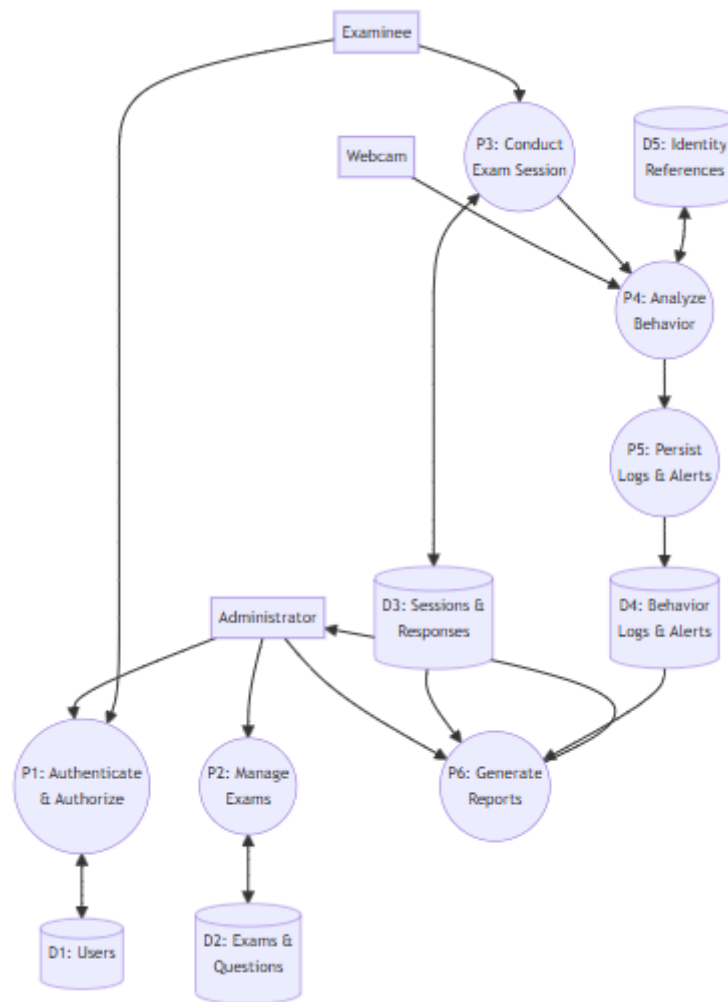


Figure #: ERD Level 1

System Design

This section discusses the planned system design, including the database structure and relationships of the proposed system.

The researchers have used the Entity Relationship Diagram (ERD) to show the logical structure and relationships of the system database. Figure 12 presents the

entities for users, examinations, questions, exam sessions, responses, session logs, behavior logs, behavior alerts, and identity references.

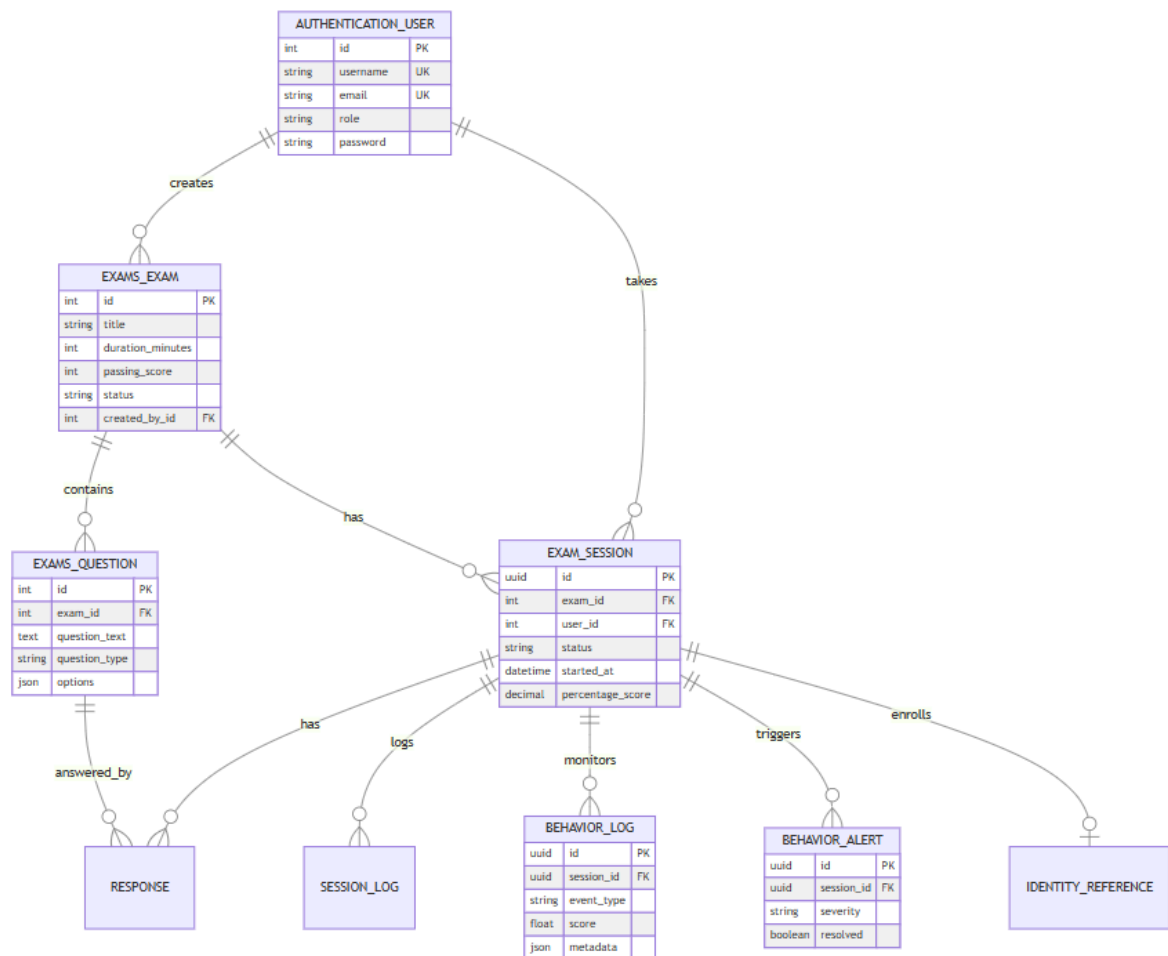


Figure #: Entity Relationship Diagram

JSON Scheme Diagram

The researchers have used the JSON Scheme Diagram to show the format of behavior-monitoring data produced by the system. The diagram includes compliance metrics, detected events, and generated alerts.

```

{
  "metrics": { "face_presence_pct", "gaze_focus_pct", "posture_compliance_pct",
               "identity_match_pct", "object_clear_pct", "overall_compliance_pct" },
  "events": [{ "event_type", "score_pct", "confidence_pct", "metadata" }],
  "alerts": [{ "type", "severity", "message", "metric_pct" }]
}

```

Figure #: Scheme Diagram

Data Dictionary

The researchers have used the Data Dictionary to provide a complete listing of the primary tables, fields, data types, constraints, and descriptions used in the Knowing Eye Examination Platform. It defines the data stored for user accounts, examinations, questions, exam sessions, responses, monitoring logs, alerts, and identity references.

Table 1. Authentication_user

Field	Type	Constraints	Description
id	INTEGER	PK,auto	Unique user identifier
username	VARCHAR(150)	UNIQUE, NOT NULL	Login username
email	VARCHAR(254)	UNIQUE, NOT NULL	Contact email
password	VARCHAR(128)	NOT NULL	Hashed password (PBKDF2)
first_name, last_name	VARCHAR(150)		Display name
role	VARCHAR(20)	ADMIN EXAMINEE	RBAC role
avatar	VARCHAR(100)	NULL	Profile image path

phone, institution, student_id	VARCHAR		Profile metadata
is_active	BOOLEAN	DEFAULT true	Account enabled flag
last_seen_at	DATETIME	NULL	Last activity timestamp
created_at, updated_at	DATETIME	auto	Audit timestamps

Table 2. Exams_exam

Field	Type	Constraints	Description
id	INTEGER	PK	Exam identifier
title	VARCHAR(255)	NOT NULL	Exam name
description, instructions	TEXT		Exam narrative and rules
duration_minutes	INTEGER	DEFAULT 120	Time limit
passing_score	INTEGER	0–100	Minimum % to pass
exam_code	VARCHAR(32)	UNIQUE, NULL	Optional access code
available_from, available_until	DATETIME	NULL	Scheduling window
max_attempts	INTEGER	DEFAULT 1	Attempts per user
status	VARCHAR(20)	draft active archived	Lifecycle state
created_by_id	INTEGER	FK → user	Authoring administrator

Table 3. exams_question

Field	Type	Constraints	Description
id	INTEGER	PK	Question identifier
exam_id	INTEGER	FK → exam	Parent exam
question_text	TEXT	NOT NULL	Prompt text
question_type	VARCHAR(20)	MCQ, T/F, etc.	Answer format
options	JSON	DEFAULT []	MCQ choices
correct_answer	TEXT	NOT NULL	Scoring key
points, order	INTEGER		Weight and display order

Table 4. Exam_sessions

Field	Type	Constraints	Description
id	UUID	PK	Session identifier (used in WebSocket URL)
exam_id, user_id	INTEGER	FK	Exam and examinee
status	VARCHAR(20)	in_progress completed terminated expired	Session state
started_at, submitted_at	DATETIME		Session timeline
time_remaining	INTEGER		Seconds left at submit
total_score, percentage_score	INT / DECIMAL	NULL until submit	Computed results

passed	BOOLEAN	NULL until submit	Pass/fail flag
ip_address, user_agent	VARCHAR / TEXT		Client audit trail

Table 5. Response, session_logs, behavior_alert, monitoring_identity_reference

Table	Key Fields	Description
responses	session_id, question_id, answer_text, is_correct, time_spent	Per-question answers submitted during session
session_logs	session_id, event_type, details, timestamp	Lifecycle audit (started, submitted, terminated)
behavior_log	session_id, event_type, score, confidence, metadata	AI-detected behavioral events
behavior_alert	session_id, alert_type, severity, resolved	Flagged integrity violations
monitoring_identity_reference	session_id (1:1), embedding (JSON), backend	Enrolled face vector for identity verification

Technologies, Concepts, and Theories

The following were the technologies, concepts, and theories applied by the researchers to develop the Knowing Eye Examination Platform. These include (wala pa)

Visual Studio

Cursor

System Testing

This section of the document presents the system testing and implementation phase of the Knowing Eye Examination Platform. The testing activities verified the functionality of user authentication, examination management, exam sessions, behavior monitoring, alerts, reports, and the artificial intelligence pipeline for facial and postural analysis.

Testing Results

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Summary of Findings

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System Maintenance

This section discusses the security and maintenance procedures of the Knowing Eye Examination Platform. These procedures ensure that the system remains secure, functional, and reliable after deployment.

Systems Security Plan

The researchers will implement security measures to protect user accounts, examination records, and behavior-monitoring data. The system will use password encryption, user authentication, role-based access control, secure HTTPS and WebSocket communication, and restricted access to examination and monitoring records. Regular database backups and security updates will also be conducted to prevent data loss and unauthorized access.

Systems Maintenance Plan

The researchers will conduct regular maintenance to identify and resolve system errors, bugs, and performance issues. The maintenance activities will include database backups, software updates, security checks, log monitoring, and testing of the examination and behavior-monitoring features. The system administrators will also update examination content, user accounts, and monitoring settings when necessary.

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